

Section 1 - Identification

Product Name
MAS® DOA Total Controls
Product Code
CDFDOAT-1, CDFDOAT-2, CDFDOAT-3, CDFDOAT-4, CDFDOAT-5, CDFDOAT-6
Address

 ThermoFisher Scientific Australia Pty Ltd
 5 Caribbean Drive, Scoresby
 VICTORIA 3179, Australia

Emergency Tel.
CHEMTREC®
03 9757 4559 or +613 9757 4559
Telephone / Fax Numbers

Tel: 1300 735 292

Fax: 1800 067 639

E-mail address

ANZinfo@thermofisher.com

Recommended Use

Laboratory chemicals.

Uses advised against

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product contains one or more substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

Section 2 - Hazard(s) Identification

Classification under Safe Work Australia

Classified as not hazardous according to criteria of Safe Work Australia.

Physical hazards

No hazards identified

Health hazards

No hazards identified

Environmental hazards

No hazards identified

Label Elements

None required

Other information

This material is prepared from a human source base. Donors have been tested by FDA approved methods and found negative for antibodies to HIV-1 and HIV-2, non-reactive for HBsAg, and non-reactive for HCV. Handle as potentially infectious material. This product does not contain any known or suspected endocrine disruptors.

Section 3 - Composition and Information on Ingredients

Component	CAS No	Weight %
Human Urine	NA	>5
Bovine Serum Albumin	9048-46-8	1-2
4-Morpholineethanesulfonic acid	4432-31-9	0.5
Sodium chloride	7647-14-5	0.45
Ethyl alcohol	64-17-5	0.3
Sodium azide	26628-22-8	<0.09
Secobarbital sodium	309-43-3	0.0002
Benzeneethanamine, N,.alpha.-dimethyl-, hydrochloride, (S)-	51-57-0	0.0002
Oxazepam	604-75-1	0.0001
1-Propanamine, 3-(10,11-dihydro-5H-dibenzo[a,d]cyclohepten-5-ylidene)-N-methyl-, hydrochloride	894-71-3	0.0001
Oxycodone hydrochloride	124-90-3	0.00008
Morphine hydrochloride	6055-06-7	0.00008
Methaqualone hydrochloride	340-56-7	0.00008
Methadone hydrochloride	1095-90-5	0.00008
Benzeneethanol, .alpha.-[2-(dimethylamino)-1-methylethyl]-.alpha.-phenyl-, propanoate (ester), hydrochloride, [S-(R*,S*)]-	1639-60-7	0.00008
Cotinine	486-56-6	0.00006
Buprenorphine	52485-79-7	0.00006
o-Benzoylecgonine	519-09-5	0.00005
Nitrazepam	146-22-5	0.00004
Lysergide	50-37-3	<0.0000003
Piperidine, 1-(1-phenylcyclohexyl)-, hydrochloride	956-90-1	0.00001

Section 4 - First Aid Measures

Inhalation	Remove to fresh air. Get medical attention immediately if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Self-Protection of the First Aider	No special precautions required.
First Aid Facilities	Eyewash, safety shower and washroom.
Most important symptoms and effects	None reasonably foreseeable.
Notes to Physician	Treat symptomatically.

Section 5 - Fire Fighting Measures

Suitable Extinguishing Media

Water spray, carbon dioxide (CO₂), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

Section 6 - Accidental Release Measures

Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

Methods for Containment and Clean Up**Clean-up methods - small spillage**

Sweep up and shovel into suitable containers for disposal.

Clean-up methods - large spillage

Not applicable, packaged goods.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

Section 7 - Handling and Storage

Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

Section 8 - Exposure Controls and Personal Protection

Exposure limits

DE - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

AUS - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

ACGIH - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

UK - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

NZ - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

Component	Australia	New Zealand WEL	ACGIH TLV	The United Kingdom	Germany
Ethyl alcohol	TWA: 1000 ppm TWA: 1880 mg/m ³	TWA: 200 ppm TWA: 380 mg/m ³ STEL: 800 ppm STEL: 1520 mg/m ³	STEL: 1000 ppm	TWA: 1000 ppm TWA; 1920 mg/m ³ TWA WEL - STEL: 3000 ppm STEL: 5760 mg/m ³ STEL	200 ppm TWA MAK; 380 mg/m ³ TWA MAK
Sodium azide	CL 0.11 ppm (0.3	Ceiling: 0.11 ppm	Ceiling: 0.29 mg/m ³	Skin	MAK 0.2 mg/m ³

	mg/m ³)	Ceiling: 0.29 mg/m ³	Ceiling: 0.11 ppm	TWA 0.1 mg/m ³ STEL 0.3 mg/m ³	(inhalable)
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Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Exposure Controls**Engineering Measures**

None under normal use conditions.

Personal protective equipment**Eye Protection**

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	AUS/NZ Standard	Glove comments
Disposable gloves	See manufacturers recommendations	-	AS/NZS 2161	(minimum requirement)

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Skin and body protection

Long sleeved clothing

Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

Recommended Filter type:

Particle filter (or AUS/NZ equivalent)

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls

No information available.

Section 9 - Physical and Chemical Properties

Information on basic physical and chemical properties

Appearance	Clear	
Physical State	Liquid	
Odor	No information available	
Odor Threshold	No data available	
pH	5	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	Not applicable	
Flash Point	Not applicable	Method - No information available
Evaporation Rate	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	

Vapor Pressure	No data available	
Vapor Density	No data available	(Air = 1.0)
Specific Gravity / Density	No data available	
Bulk Density	Not applicable	Liquid
Water Solubility	Miscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
4-Morpholineethanesulfonic acid	-2.2	
Ethyl alcohol	-0.32	
Secobarbital sodium	1.97	
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
Viscosity	No data available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Other information**Section 10 - Stability and Reactivity**

Reactivity	None known, based on information available
Stability	Stable under normal conditions.
Conditions to Avoid	Incompatible products, Excess heat.
Incompatible Materials	None known.
Hazardous Decomposition Products	None under normal use conditions.
Hazardous Polymerization	Hazardous polymerization does not occur.

Section 11 - Toxicological Information**Information on Toxicological Effects****Product Information****(a) acute toxicity;****Oral**

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Sodium chloride	LD50 = 3550 mg/kg (Rat)	LD50 > 10000 mg/kg (Rabbit)	LC50 > 42 mg/L (Rat) 1 h
Ethyl alcohol	LD50 = 7060 mg/kg (Rat)		20000 ppm/10H (Rat)
Sodium azide	LD50 = 27 mg/kg (Rat)	-	LC50 0.054 - 0.52 mg/L (Rat) 4 h
Secobarbital sodium	LD50 = 125 mg/kg (Rat)		
Oxazepam	LD50 > 8 g/kg (Rat)		
1-Propanamine, 3-(10,11-dihydro-5H-dibenzo[a,d]cyclohept	LD50 = 405 mg/kg (Rat)		

en-5-ylidene)-N-methyl-, hydrochloride			
Morphine hydrochloride	LD50 = 335 mg/kg (Rat)		
Methaqualone hydrochloride	LD50 = 240 mg/kg (Rat)		
Methadone hydrochloride	LD50 = 30 mg/kg (Rat)		
Benzeneethanol, alpha.-[2-(dimethylamino)-1-methylethyl]-.a lpha.-phenyl-, propanoate (ester), hydrochloride, [S-(R*,S*)]-	LD50 = 84 mg/kg (Rat)		
Nitrazepam	LD50 = 825 mg/kg (Rat)		
Piperidine, 1-(1-phenylcyclohexyl)-, hydrochloride	LD50 = 135 mg/kg (Rat)		

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available

Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

Component	Australia	New Zealand	New South Wales	Western Australia	IARC	EU	UK	Germany
Oxazepam					Group 2B			

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available

Section 12 - Ecological Information

Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Sodium chloride	Pimephals prome: LC50: 7650 mg/L/96h	EC50: 1000 mg/L/48h		
Ethyl alcohol	Fathead minnow (Pimephales promelas) LC50 = 14200 mg/L/96h	EC50 = 9268 mg/L/48h EC50 = 10800 mg/L/24h	EC50 (72h) = 275 mg/l (Chlorella vulgaris)	Photobacterium phosphoreum: EC50 = 34634 mg/L/30 min Photobacterium phosphoreum: EC50 = 35470 mg/L/5 min
Sodium azide	LC50: = 0.7 mg/L, 96h			

	(Lepomis macrochirus) LC50: = 0.8 mg/L, 96h (Oncorhynchus mykiss) LC50: = 5.46 mg/L, 96h flow-through (Pimephales promelas)			
Secobarbital sodium	LC50: 21.4 - 26 mg/L, 96h flow-through (Pimephales promelas)			

Persistence and Degradability**Persistence**

Miscible with water, Persistence is unlikely, based on information available.

Bioaccumulative Potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
4-Morpholineethanesulfonic acid	-2.2	No data available
Ethyl alcohol	-0.32	No data available
Secobarbital sodium	1.97	No data available

Mobility

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility Highly mobile in soils

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

Persistent Organic Pollutant

This product does not contain any known or suspected substance

Ozone Depletion Potential

This product does not contain any known or suspected substance

Section 13 - Disposal Considerations

Waste from Residues/Unused Products

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

Contaminated Packaging

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

Other Information

Chemical wastes should be disposed through a licensed commercial waste collection service.

Section 14 - Transport Information

IMDG/IMO

Not regulated

ADG

Not regulated

Component	Hazchem Code
Ethyl alcohol 64-17-5 (0.3)	2YE 2Y
Sodium azide 26628-22-8 (<0.09)	2XE

IATA

Not regulated

Environmental hazards

No hazards identified

Special Precautions

No special precautions required

Additional information

None known

Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

National Regulations

Australia

See section 8 for national exposure control parameters.

Standard for the Uniform Scheduling of Medicines and Poisons

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

Component	Standard for the Uniform Scheduling of Medicines and Poisons
Secobarbital sodium - 309-43-3	Schedule 4 listed - except when separately specified in these Schedules Schedule 8 listed
Oxazepam - 604-75-1	Schedule 4 listed - present
Buprenorphine - 52485-79-7	Schedule 8 listed
Nitrazepam - 146-22-5	Schedule 4 listed - present
Lysergide - 50-37-3	Schedule 9 listed

Australian Industrial Chemicals Introduction Scheme (AICIS)

Component	Australian Industrial Chemicals Introduction Scheme (AICIS)	Additional information
Bovine Serum Albumin - 9048-46-8	Present	-
4-Morpholineethanesulfonic acid - 4432-31-9	Present	-
Sodium chloride - 7647-14-5	Present	-
Ethyl alcohol - 64-17-5	Present	-
Sodium azide - 26628-22-8	Present	-

Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

Chemicals of Security Concern

This product contains one or more substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

Component	Australian - Illicit Drug Precursors/Reagents Substance List	Chemicals of Security Concern
Sodium azide - 26628-22-8		Listed in Appendix A Precursors to homemade explosives - concentration >=95%

Legend

Chemicals of Security Concern - for further information see <http://www.chemicalsecurity.gov.au/securityconcerns>

National pollutant inventory

Subject to reporting requirements

Component	National pollutant inventory
Ethyl alcohol - 64-17-5	10 tonne/yr. Threshold category 1

Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

International Inventories

Component	AICS	NZIoC	EINECS	ELINCS	TSCA	DSL	NDSL	PICCS	ENCS	ISHL	IECSC	KECL
Bovine Serum Albumin	X	X	232-936-2	-	X	X	-	X	-		X	KE-05-0011
4-Morpholineethanesulfonic acid	X	X	224-632-3	-	X	X	-	-	-	X	X	KE-25529
Sodium chloride	X	X	231-598-3	-	X	X	-	X	X	X	X	KE-31387
Ethyl alcohol	X	X	200-578-6	-	X	X	-	X	X	X	X	KE-13217
Sodium azide	X	X	247-852-1	-	X	X	-	X	X	X	X	KE-31357
Secobarbital sodium	-	X	206-218-4	-	-	X	-	-	X	X	-	-
Benzeneethanamine, N,.alpha.-dimethyl-, hydrochloride, (S)-	-	X	200-106-9	-	X	-	X	-	-		-	-
Oxazepam	-	X	210-076-9	-	-	X	-	-	X	X	X	-
1-Propanamine, 3-(10,11-dihydro-5H-dibenzo[a,d]cyclohepten-5-ylidene)-N-methyl-, hydrochloride	-	X	212-973-0	-	-	X	-	X	-		-	-
Oxycodone hydrochloride	-	-	204-717-1	-	-	X	-	-	-		-	-
Methaqualone hydrochloride	-	X	206-431-2	-	-	-	-	-	-		-	-
Methadone hydrochloride	-	X	214-140-7	-	-	-	-	-	-		-	-
Benzeneethanol, .alpha.-[2-(dimethylamino)-1-methylethyl]-.alpha.-phenyl-, propanoate (ester), hydrochloride, [S-(R*,S*)]-	-	X	216-683-5	-	X	X	-	-	-		-	-
Cotinine	-	-	207-634-9	-	-	-	-	-	-		-	-
Buprenorphine	-	X	257-950-6	-	-	-	-	-	-		-	-
o-Benzoylcegonine	-	-	208-263-5	-	-	-	-	-	-		-	-
Nitrazepam	-	X	205-665-2	-	-	X	-	-	X	X	-	KE-10723
Lysergide	-	-	200-033-2	-	-	-	-	-	-		-	-
Piperidine, 1-(1-phenylcyclohexyl)-, hydrochloride	-	X	-	-	-	-	-	-	-		-	-

Legend: X - Listed, '-' - Not Listed, XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B)). **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

International Regulations

Ozone Depletion Potential This product does not contain any known or suspected substance

Persistent Organic Pollutant This product does not contain any known or suspected substance

Rotterdam Convention (PIC) Not applicable

Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

Component	Basel Convention (Hazardous Waste)	Australian Hazardous Waste Act - Categories of Wastes to Be Controlled
Ethyl alcohol - 64-17-5	Annex I - Y42	Y42 except Halogenated solvents

Component	CAS No	OECD HPV	Restriction of Hazardous Substances (RoHS)	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Human Urine	NA	Not applicable	Not applicable	Not applicable	Not applicable
Bovine Serum Albumin	9048-46-8	Not applicable	Not applicable	Not applicable	Not applicable
4-Morpholineethanesulfonic acid	4432-31-9	Not applicable	Not applicable	Not applicable	Not applicable
Sodium chloride	7647-14-5	Listed	Not applicable	Not applicable	Not applicable
Ethyl alcohol	64-17-5	Listed	Not applicable	Not applicable	Not applicable
Sodium azide	26628-22-8	Not applicable	Not applicable	Not applicable	Not applicable
Secobarbital sodium	309-43-3	Not applicable	Not applicable	Not applicable	Not applicable
Benzeneethanamine, N,.alpha.-dimethyl-, hydrochloride, (S)-	51-57-0	Not applicable	Not applicable	Not applicable	Not applicable
Oxazepam	604-75-1	Not applicable	Not applicable	Not applicable	Not applicable
1-Propanamine, 3-(10,11-dihydro-5H-dibenzo[a,d]cyclohepten-5-ylidene)-N-methyl-, hydrochloride	894-71-3	Not applicable	Not applicable	Not applicable	Not applicable
Oxycodone hydrochloride	124-90-3	Not applicable	Not applicable	Not applicable	Not applicable
Morphine hydrochloride	6055-06-7	Not applicable	Not applicable	Not applicable	Not applicable
Methaqualone hydrochloride	340-56-7	Not applicable	Not applicable	Not applicable	Not applicable
Methadone hydrochloride	1095-90-5	Not applicable	Not applicable	Not applicable	Not applicable
Benzeneethanol, .alpha.-[2-(dimethylamino)-1-methylethyl]-.alpha.-phenyl-, propanoate (ester), hydrochloride, [S-(R*,S*)]-	1639-60-7	Not applicable	Not applicable	Not applicable	Not applicable
Cotinine	486-56-6	Not applicable	Not applicable	Not applicable	Not applicable
Buprenorphine	52485-79-7	Not applicable	Not applicable	Not applicable	Not applicable
o-Benzoyllecgonine	519-09-5	Not applicable	Not applicable	Not applicable	Not applicable
Nitrazepam	146-22-5	Not applicable	Not applicable	Not applicable	Not applicable
Lysergide	50-37-3	Not applicable	Not applicable	Not applicable	Not applicable
Piperidine, 1-(1-phenylcyclohexyl)-, hydrochloride	956-90-1	Not applicable	Not applicable	Not applicable	Not applicable

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Secobarbital sodium	-	Use restricted. See entry 75. (see link for restriction details)	-
Oxazepam	-	Use restricted. See entry 75. (see link for restriction details)	-
Benzeneethanol, .alpha.-[2-(dimethylamino)-1-methylethyl]-.alpha.-phenyl-, propanoate (ester), hydrochloride, [S-(R*,S*)]-	-	Use restricted. See entry 75. (see link for restriction details)	-
Nitrazepam	-	Use restricted. See entry 75. (see link for restriction details)	-
Lysergide	-	Use restricted. See entry 75. (see link for restriction details)	-

<https://echa.europa.eu/substances-restricted-under-reach>

Section 16 - Other Information

Legend

AICS - Australian Inventory of Chemical Substances	NZIoC - New Zealand Inventory of Chemicals
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory	EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List	ENCS - Japanese Existing and New Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances	KECL - Korean Existing and Evaluated Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances	CAS - Chemical Abstracts Service
TWA - Time Weighted Average	ACGIH - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)
IARC - International Agency for Research on Cancer	IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association	ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail
MARPOL - International Convention for the Prevention of Pollution from Ships	OECD - Organisation for Economic Co-operation and Development
NZS 5433:2020 - Transport of Dangerous Goods on Land	LC50 - Lethal Concentration 50%
LD50 - Lethal Dose 50%	ATE - Acute Toxicity Estimate
EC50 - Effective Concentration 50%	RPE - Respiratory Protective Equipment
WEL - Workplace Exposure Limit	NOEC - No Observed Effect Concentration
DNEL - Derived No Effect Level	BCF - Bioconcentration factor
POW - Partition coefficient Octanol:Water	PBT - Persistent, Bioaccumulative, Toxic
vPvB - very Persistent, very Bioaccumulative	
VOC - (Volatile Organic Compound)	

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Revision Date	12-Mar-2025
Revision Summary	Update to GHS format.

This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet